KALAM (கலம்) — AGRICULTURAL INSIGHTS ATLAS

Ambasamudram & Cheranmahadevi Taluks, Tirunelveli · Multi-Year EO Analytics (2023–2026)

CROPPING INTENSITY

23.0% Double-Cropped Cropland

CANOPY VIGOUR

98.6% Normal / Vigorous Cropland

ANNUAL WATER DEMAND

42.64 Million m³ (MCM) Gross Requirement

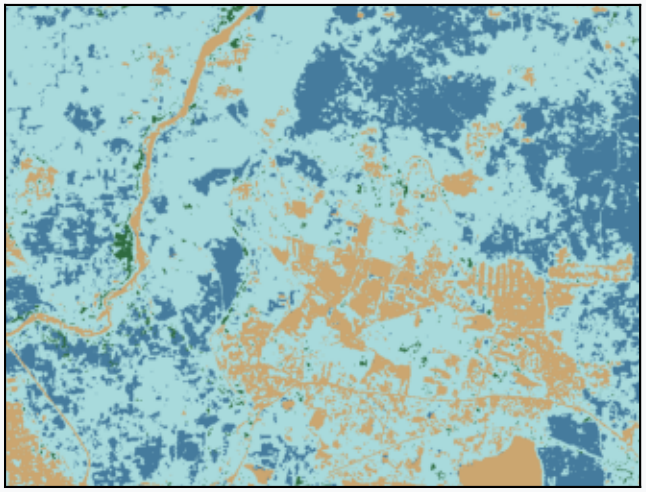
EXECUTIVE INSIGHT SUMMARY & POLICY ACTION ITEMS

- **Cropping Intensity:** 23.0% of cropland (693.2 ha) is double-cropped with Kar and Pishanam paddy cycles, with 1.3% under perennial banana.
- **Transplanting Calendar:** Peak Kar transplanting occurs in Jul (55.6% of Kar fields), while main Pishanam sowing peaks in Nov (76.6% of Pishanam fields). 0.0% of Pishanam acreage experiences delayed transplanting beyond November.
- **Irrigation Dependency & Water Access:** 51.3% of paddy acreage (886.3 ha) is concentrated within 500 m of river channels, canals, or system irrigation tanks, and 79.6% is located within 1 km. Perennial banana comprises 3.4% of cropland in the immediate 0–250 m corridor.
- **Crop Change & Farmland Dynamics:** Evaluated 3,011.2 ha of 3,011.2 ha cropland (confidence ≥ 0.60): 52.9% of paddy and 1.3% of banana parcels remained stable across 2023–2026, with 100.5 ha (3.3%) identified as candidate paddy-to-banana diversification.
- **Crop Health & NDVI Anomaly:** 98.6% of analyzed cropland displays normal or above-normal vegetative vigour. Vegetative moisture/vigour stress is observed across 1.4% (35.6 ha) of cultivated area.
- **Indicative Gross Irrigation Requirement:** Total indicative gross irrigation requirement across analyzed farmed plains is 42.6 MCM/year. Paddy cycles (Kar & Pishanam) account for 88.5% (37.7 MCM), while perennial banana requires 0.8 MCM (1.9%).

Kalam © 2026 · Problem Statement 1 · SustainAI / Mapathon · Contains modified Copernicus Sentinel data

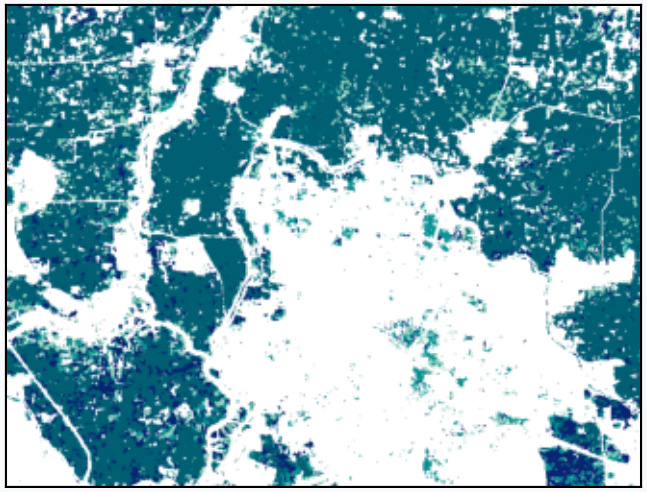
PHENOLOGY DYNAMICS & IRRIGATION ACCESS

01. Cropping Intensity



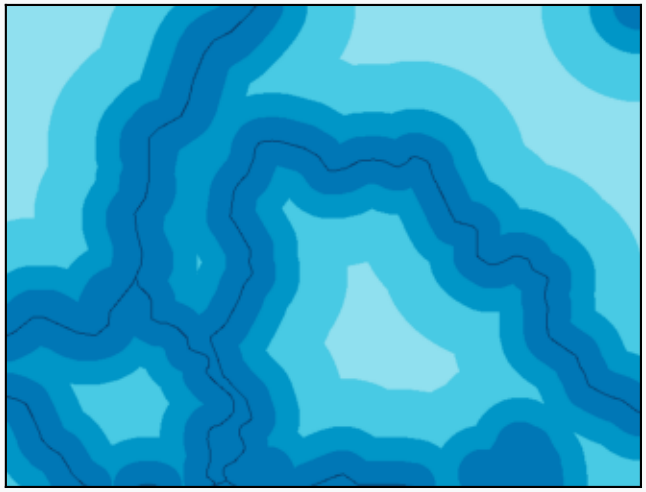
ltivated (144.3 ha, 4.8%) | Single Crop (2133.3 ha, 70.8%) | Double Crop (693.2 ha, 23.0%)

02. Transplanting Calendar



Kar Transplanted (0 ha, 0.0%) | Pishanam Peak (Oct) | Pishanam Late (Nov–Dec)

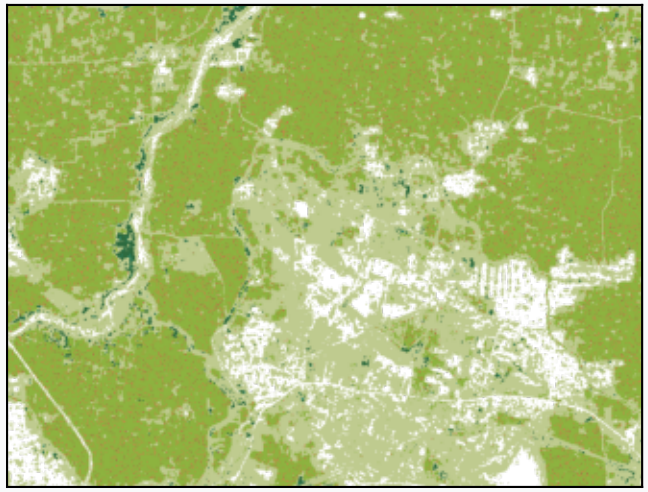
03. Irrigation Dependency



0 – 250 m | 250 – 500 m | 500 – 1,000 m | 1,000 – 2,000 m

FARMLAND DYNAMICS, VIGOUR & IRRIGATION DEMAND

04. Crop Change & Loss



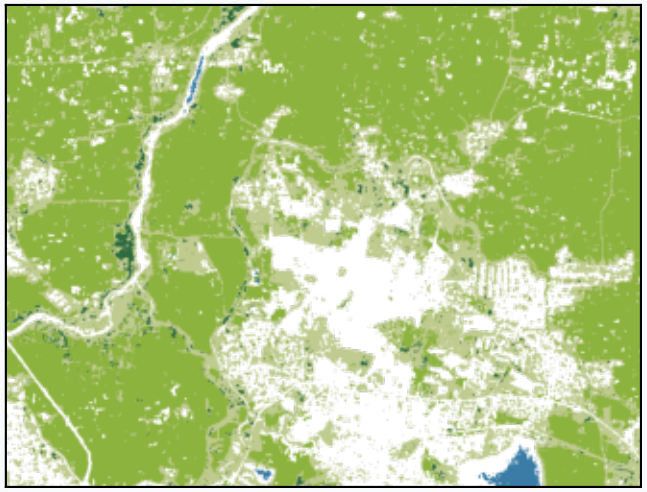
Stable Paddy | Stable Banana | Candidate Paddy → Banana | Severe Stress (0.3%) | Stressed Paddy (1.1%) | Stressed Banana (0.3%)

05. NDVI Vigour Anomaly



Normal Vigour (2546.29 sq.km, 98.4%) | Above Normal (3.90 sq.km, 0.1%)

06. Water Demand Budget



Kar Paddy (Short-duration) (0.70 MCM, 1.6%) | Paddy (Main season) (20.74 MCM, 48.6%) | Perennial Banana (Year-round) (0.81 MCM, 1.9%